

Assignment 5 RM: Counterparty Credit Risk

Group 12

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1 Proof of the affine trick

The short rate in the Hull–White extended Vasicek model is

$$r_t = \varphi(t) + x_t, \quad (1)$$

where $\varphi(t)$ is a deterministic function of time and x_t satisfies the Ornstein–Uhlenbeck SDE

$$\begin{cases} dx_t = -a x_t dt + \sigma dW_t, \\ x_{t_0} = x_0. \end{cases} \quad (2)$$

Of which the solution is well known :

$$x_t = x_0 e^{-a(t-t_0)} + \sigma \int_{t_0}^t e^{-a(t-u)} dW_u \quad (3)$$

The forward discount factor ratio between maturities T_1 and T_2 , observed at time t , satisfies (Lemma 1):

$$\frac{B(t, T_1, T_2)}{B(t_0, T_1, T_2)} = \exp \left(-x_t \left(\frac{\sigma(t, T_2) - \sigma(t, T_1)}{\sigma} \right) - \frac{1}{2} \int_{t_0}^t [\sigma^2(u, T_2) - \sigma^2(u, T_1)] du \right), \quad (4)$$

with

$$\sigma(u, \tau) = \sigma \cdot \frac{1 - e^{-a(\tau-u)}}{a} \quad (5)$$

To recover the *spot* discount factor $B(s, \tau)$ we set $T_1 = t_0$ and $T_2 = \tau$ in (3), evaluated at $t = s$. Since $\sigma(u, t_0) = 0$ for all $u \geq t_0$, three simplifications occur:

- $B(t_0, t_0, \tau) = B(t_0, \tau)$ and $B(t_0, t_0, s) = 1$, so $B(t_0, T_1, T_2) = B(t_0, \tau)$;
- $B(s, t_0, \tau) = B(s, \tau)$ and $B(s, t_0, s) = 1$, so $B(s, T_1, T_2) = B(s, \tau)$;
- in the exponent, $\sigma(s, T_1)/\sigma = 0$ and $\sigma(u, T_1)^2 = 0$.

Substituting into (3):

$$\frac{B(s, \tau)}{B(t_0, \tau)} = \exp \left(-x_s \frac{\sigma(s, \tau)}{\sigma} - \frac{1}{2} \int_{t_0}^s \sigma(u, \tau)^2 du \right). \quad (6)$$

Multiplying both sides by $B(t_0, \tau)$ and dividing by $B(t_0, s)$ (to normalise the discount factor curve at s):

$$B(s, \tau) = \underbrace{\frac{B(t_0, \tau)}{B(t_0, s)} \exp \left(-\frac{1}{2} \int_{t_0}^s [\sigma(u, \tau)^2 - \sigma(u, s)^2] du \right)}_{A(s, \tau)} \cdot \exp(-x_s C(s, \tau)), \quad (7)$$

where

$$C(s, \tau) = \frac{\sigma(s, \tau)}{\sigma} = \frac{1 - e^{-a(\tau-s)}}{a}.$$

The $\sigma(u, s)^2$ term in the integral of A arises from the normalisation by $B(t_0, s)$, which introduces the same exponential structure evaluated at maturity s . Q.E.D

2 → 5 Base case

Polimi Bank enters a 10y payer IRS on Feb 15, 2008, with notional $N = 100\text{M}$. Fixed leg: annual, 30/360 European. Floating leg: 3M Euribor, ACT/360. The fixed rate is set at par.

Model. The short rate follows the Hull–White extended Vasicek model with $a = 0.0205$ and $\sigma = 0.0094$. We simulate x_t on a quarterly grid (40 dates, 10 years) for 250,000 scenarios. At each simulation date s we recover the full discount curve via the affine trick proven in Section 1.

MtM. For a payer IRS at par, the mark-to-future viewed from Polimi Bank is

$$\text{MtM}(s) = N \cdot \left[\underbrace{1 - B(s, T_N)}_{\text{floating leg}} - K \underbrace{\sum_{T_i > s} \tau_i B(s, T_i)}_{\text{fixed leg}} \right]. \quad (8)$$

Past cash flows are excluded with a mask on $T_i > s$. Each simulation date is also a reset date of the floating leg, so the floating leg value reduces to $1 - B(s, T_N)$.

Exposure profiles. We compute the Expected Positive Exposure

$$\text{EPE}(s) = \mathbb{E}[\max(\text{MtM}(s), 0)]$$

and the 95% Potential Future Exposure

$$\text{PFE}_{95\%}(s) = Q_{95\%}[\max(\text{MtM}(s), 0)].$$

Both profiles show the classic "hump" shape typical of a payer IRS. The exposure grows until roughly the middle of the swap life, then decays as fewer cash flows remain.

Peak-PFE.

	Peak date	Peak-PFE
Base case	2012-02-20	16,636,540

The peak materialises around 4 years from inception, slightly less than half of the maturity. This is the standard signature of an at-par payer swap: as rates diffuse, MtM moves both ways, but the residual annuity is still large enough to amplify the upside.

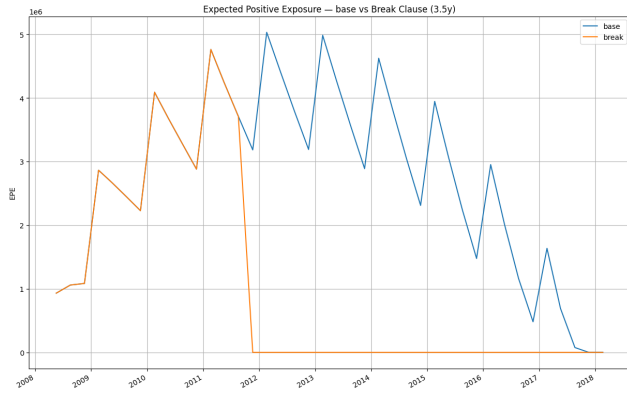
6.a Break Clause

A mandatory break at $T_{\text{br}} = 3.5\text{y}$ forces both parties to cash-settle the swap at its market value on Aug 19, 2011. By risk-neutral pricing (tower property of the conditional expectation), the value of the contract before the break is identical to the value of the original 10y swap:

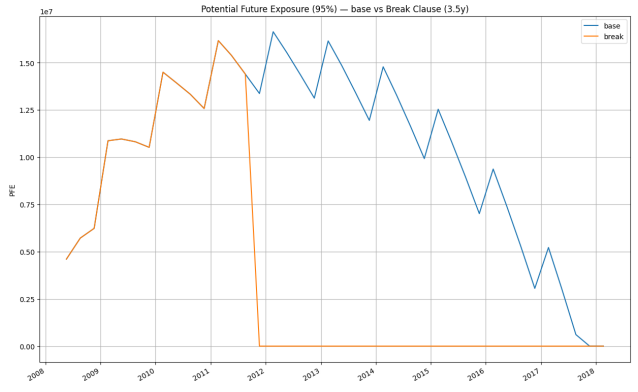
$$\text{MtM}^{\text{br}}(s) = \begin{cases} \text{MtM}(s) & s < T_{\text{br}}, \\ 0 & s \geq T_{\text{br}}. \end{cases} \quad (9)$$

The contract is therefore equivalent to the base swap up to T_{br} and has no counterparty exposure thereafter.

Result.



(a) EPE — base vs break clause.



(b) 95% PFE — base vs break clause.

	Peak date	Peak-PFE
Base case	2012-02-20	16,636,540
Break (3.5y)	2011-02-21	16,166,870

The exposure profile coincides with the base case up to T_{br} and collapses to zero immediately after. The peak-PFE is therefore the maximum of the base profile restricted to $s < T_{br}$. In our simulation this happens at $s \approx 3y$, just before the break, with a value about 3% below the unconstrained peak. As a CCR mitigant, the break clause is effective only on the back-end of the profile: it removes all exposure after T_{br} but leaves the front-end of the curve unchanged.

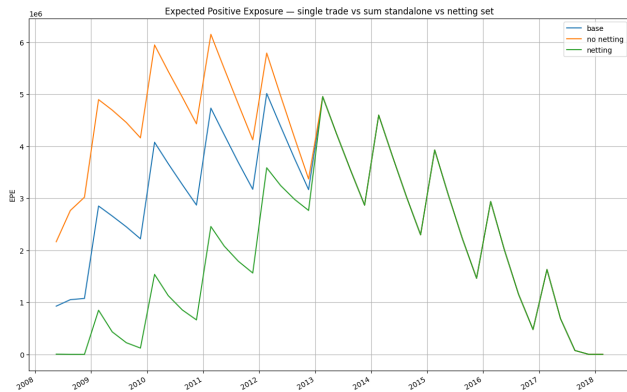
6.b Netting agreement

Second trade. A 5y receiver IRS at par, $N_2 = 150M$, both legs quarterly. Its MtM is

$$\text{MtM}_2(s) = N_2 \cdot \left[K_2 \sum_{T_j > s} \tau_j B(s, T_j) - (1 - B(s, T_N^{(2)})) \right] \quad \text{for } s \leq T_N^{(2)}. \quad (10)$$

Three measures of exposure. For the same Monte Carlo scenarios we compare:

- single trade (base 10y payer);
- no-netting (two standalone trades): $\max(\text{MtM}, 0) + \max(\text{MtM}_2, 0)$;
- netting set (single legal exposure): $\max(\text{MtM} + \text{MtM}_2, 0)$.



(a) EPE — single trade vs sum standalone vs netting set.



(b) 95% PFE — single trade vs sum standalone vs netting set.

	Peak date	Peak-PFE
Base (single trade)	2012-02-20	16,636,540
No netting (sum of two standalone)	2012-02-20	16,636,540
Netting set	2013-02-19	16,150,230

Comment. Both swaps depend on the same state variable x_t , but with opposite directional exposures (payer vs receiver). The receiver MtM tends to be negative on the scenarios where the payer MtM is positive, so $\text{MtM} + \text{MtM}_2$ has smaller positive tails than $\max(\text{MtM}, 0) + \max(\text{MtM}_2, 0)$. This is exactly what the netting peak-PFE captures: it is below the no-netting value. The peak-PFE of the no-netting portfolio coincides with the base case because both peaks occur at $s \approx 4\text{y}$, when the 5y receiver swap has less than one year of residual life; at this date, $\max(\text{MtM}_2, 0)$ is negligible across essentially all scenarios, so the 95th percentile of the sum equals that of the payer alone. The benefit of netting is more pronounced in the inner part of the profile, where both trades are alive and partially offset each other.

6.c CSA agreement

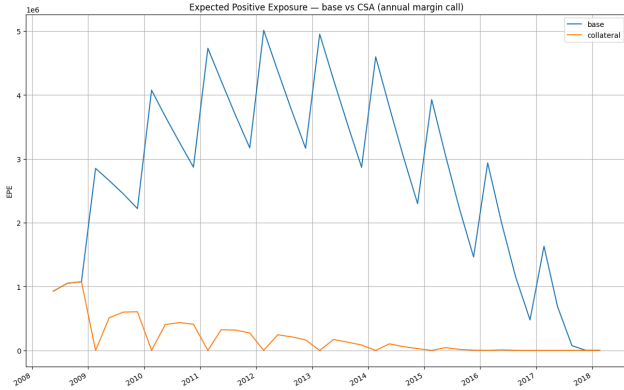
Parameters. Following the assignment requirements (Polimi never posts, the counterparty covers fully, no IA, annual margin call, no settlement lag):

	Polimi Bank (B)	Counterparty (C)
Threshold	$+\infty$	0
MTA	0	0
Cap	0	$+\infty$

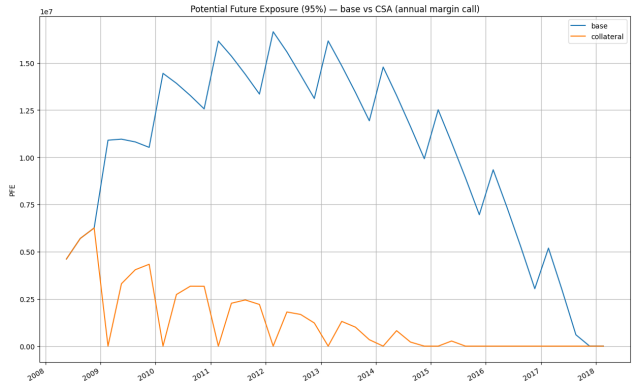
$\text{THR}_B = +\infty$ removes any obligation for B to post (the unsecured exposure C is willing to bear is unbounded). $\text{Cap}_B = 0$ enforces the same constraint from the other side. $\text{THR}_C = 0$ forces C to collateralise its entire positive exposure. $\text{Cap}_C = +\infty$ ensures no upper limit on C's posted collateral.

Mechanics. Posted collateral is remunerated at the risk-free rate, proxied by the shortest HW zero rate at each step. Margin calls happen every 12 months. The collateralised exposure is $\max(\text{MtM}(s) - \text{VM}(s), 0)$.

Result.



(a) EPE — base vs CSA.



(b) 95% PFE — base vs CSA.

	Peak date	Peak-PFE
Base case	2012-02-20	16,636,540
With CSA	2008-11-19	6,232,333

The PFE profile becomes a sawtooth: exposure builds up over the 12 months between margin calls, then resets close to zero. The peak therefore falls at 9 months, just before the first margin call. The peak-PFE is reduced by roughly $2.7\times$, and the EPE profile is essentially flat at a low level for the rest of the swap life. The CSA is the strongest mitigant of the three.

7. CVA approximation

Under the assumptions of the assignment (independence of exposure and default, B does not default, flat hazard rate $\lambda = 200$ bps, LGD = 60%), the CVA at t_0 is

$$\text{CVA}(t_0) \approx \text{LGD} \cdot \sum_{i=1}^{40} B(0, t_i) \cdot \text{EPE}(t_i) \cdot [S(t_{i-1}) - S(t_i)], \quad (11)$$

where $S(t) = e^{-\lambda t}$ is the survival probability. The term $S(t_{i-1}) - S(t_i)$ is the marginal default probability in $(t_{i-1}, t_i]$.

CVA across scenarios.

Scenario	CVA (t_0)
Base case	254,249
Break clause (3.5y)	107,064
No netting (two standalone)	325,828
Netting set	166,167
With CSA	22,253

The ranking matches the PFE intuition. The break clause divides the CVA by roughly 2.4 $\times$: by chopping off the second half of the exposure profile (which carries roughly 58% of the discounted-and-default-weighted EPE area), it produces a meaningful but moderate mitigation. Netting roughly halves the no-netting CVA. The CSA reduces the CVA by an order of magnitude, in line with the strong reduction of the exposure profile.

Sensitivities (base case). We compute three finite-difference sensitivities of the CVA, keeping the EPE profile fixed:

Risk factor	ΔCVA
+1 bp parallel shift on the discount curve (discount only)	−110
+1 bp hazard rate λ	+1,164
+1 pp on LGD	+4,237

The λ sensitivity is the strongest in relative terms: higher hazard rate means higher default probability over the 10-year horizon, which scales the CVA proportionally. The LGD sensitivity is exactly linear, since CVA is proportional to LGD. The rate sensitivity is reported in a *discount-only* approximation: a parallel shift on the curve also affects the MtM and hence the EPE of the underlying payer swap. For a payer IRS, an upward shift in rates increases the MtM, so the EPE channel dominates and the *full* CVA rate sensitivity is positive. Capturing it correctly would require re-bootstrapping the curve and re-running the Monte Carlo simulation, which is left out for brevity. The main market risk factors of the CVA are therefore the level and shape of the IR curve (through the MtM channel) and the credit spread of the counterparty (through λ).

Full rate sensitivity (re-bootstrap). To verify the discount-only approximation, we also computed the rate sensitivity of the CVA by re-bootstrapping the curve with ± 1 bp parallel shifts on the market quotes (depo, futures and swap rates) and re-running the Monte Carlo pricing of the base-case 10y payer IRS. The contract is kept at the original fixed rate K (no re-pricing at par), so the bump makes the existing trade in-the-money on average. The result is

$$\Delta\text{CVA}_{\text{full}} = +1,827 \quad \text{per } +1 \text{ bp parallel shift,}$$

positive and roughly 17 $\times$ larger in absolute value than the discount-only approximation, and of opposite sign. The decomposition $\Delta\text{CVA}_{\text{full}} = \Delta\text{CVA}_{\text{disc-only}} + (\text{EPE channel}) = -110 + 1,937$ confirms that the EPE channel — the impact of rate moves on the swap MtM and hence on the exposure profile

— dominates the discount channel by two orders of magnitude. For a payer IRS this is the correct intuition: a parallel hike in rates makes the bank receive a higher Euribor against an unchanged fixed coupon, so the swap moves in-the-money and the CVA rises.

Errors Found in the Template Code

1. **Inverted swap signs** (`ex1_utilities.py`, `swap_mtm`): the multipliers for `PAYER` and `RECEIVER` were swapped. The payer convention requires a positive multiplier on the floating–fixed difference, while the receiver requires a negative one. The error does not propagate into the simulation, where the MtM is computed manually with correct signs.
2. **Wrong date offset unit** (Notebook, MC grid setup): the simulation horizon was built with `month_offset` instead of `year_offset`, producing a 10-month grid rather than a 10-year one.
3. **Uninitialised prev_date** (`ex5_utilities.py`, `simulate_ou_process`): the variable `prev_date` was set to `None` inside the loop at every iteration and never updated, causing a crash at the first time-step computation. It must be initialised to the start date before the loop and updated to the current simulation date at the end of each iteration.
4. **Incomplete volatility function** (`ex5_utilities.py`, `affine_trick`): `sigma_func` was defined with a single argument. The correct implementation requires two arguments (u, τ) and must return $\sigma(u, \tau) = \frac{\sigma}{a}(1 - e^{-a(\tau-u)})$.